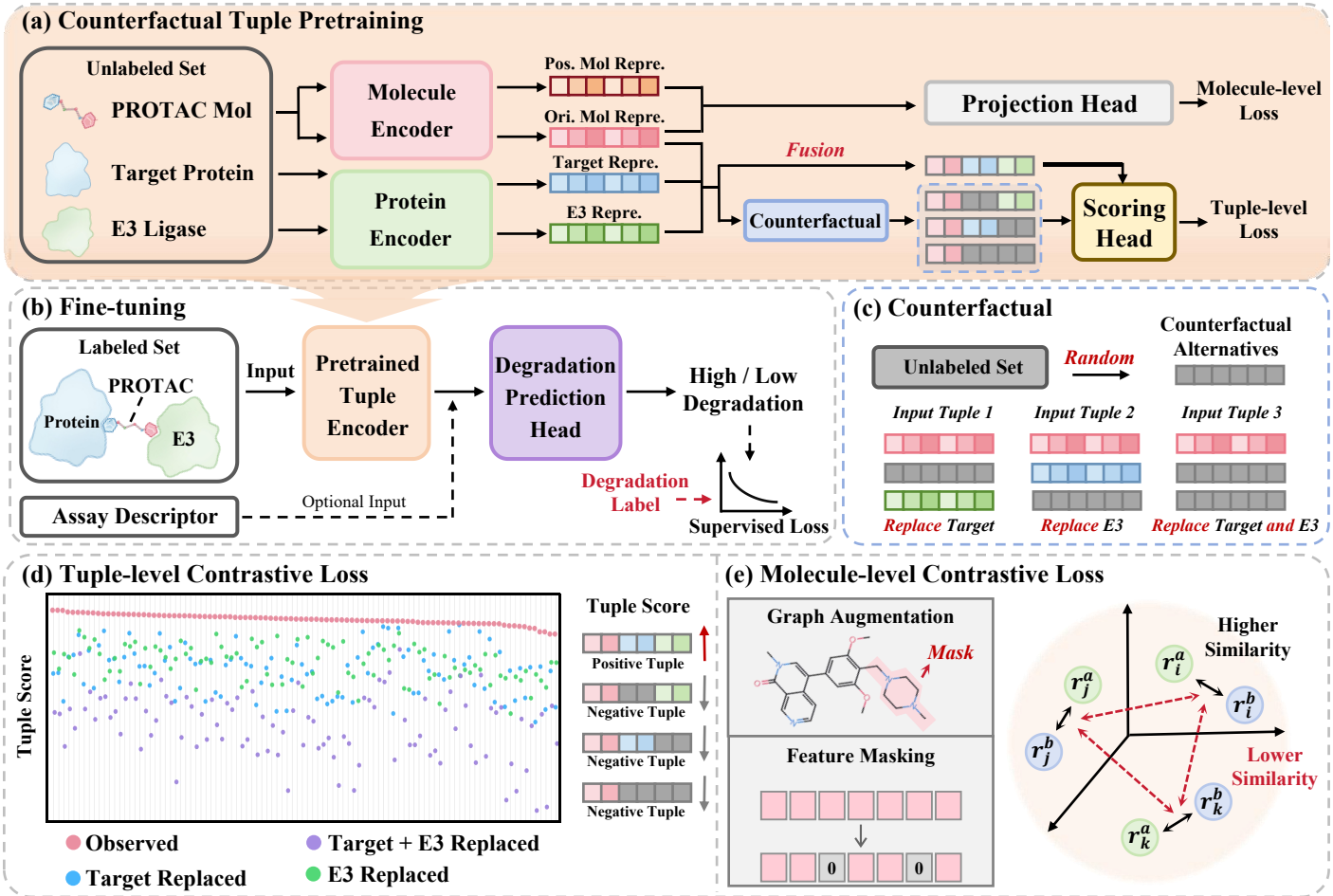

(a) Counterfactual Tuple Pretraining
Unlabeled Set
PROTAC Mol
Target Protein
E3 Ligase
Molecule Encoder
Protein Encoder
Pos. Mol Repre.
Ori. Mol Repre.
Target Repre.
E3 Repre.
Projection Head
Molecule-level Loss
Fusion
Counterfactual
Scoring Head
Tuple-level Loss
(b) Fine-tuning
Labeled Set
PROTAC
Protein
E3
Assay Descriptor
Input
Optional Input
Pretrained Tuple Encoder
Degradation Prediction Head
High / Low Degradation
Degradation Label
Supervised Loss
(c) Counterfactual
Unlabeled Set
Random
Counterfactual Alternatives
Input Tuple 1
Input Tuple 2
Input Tuple 3
Replace Target
Replace E3
Replace Target and E3
(d) Tuple-level Contrastive Loss
Tuple Score
Observed
Target Replaced
Target + E3 Replaced
E3 Replaced
Tuple Score
Positive Tuple
Negative Tuple
Negative Tuple
Negative Tuple
(e) Molecule-level Contrastive Loss
Graph Augmentation
Mask
Feature Masking
0
0
Higher Similarity
Lower Similarity